DAY 19 — AI ETHICS & LIMITATIONS

Bias Analysis, Regulations (GDPR & AI Act), Governance, Public Policy & Data Privacy

Learning outcome:

- Understand data privacy,
- Security, and confidentiality risks,
- Identify legal accountabilities,
- Safely apply AI in professional environments.

Hour 1: Real-world incidents of AI bias

Theory:

• Real-world AI bias occurs when algorithms produce systematically prejudiced decisions that harm specific demographic groups.

• Technical causes: skewed training datasets, proxy variables (e.g., zip codes reflecting race), historical labeling biases, and flawed loss functions.

• **High-profile historical cases:**

- Amazon Recruiting Tool: penalized CVs containing the word "women's".

- COMPAS Algorithm: showed higher false-positive recidivism rates for Black defendants.

- Healthcare Risk Algorithms: spent less care funding on minority patients due to past billing disparity.

Mechanisms of failure:

• Historical bias: AI learns past human discrimination as optimal behavior.

• Representation bias: minority groups are underrepresented in image or text data.

• Measurement bias: measuring wrong indicators (e.g., healthcare spending instead of actual illness).

Key lesson:

Algorithms do not understand fairness or justice; they optimize strictly on patterns found in historical data.

Role of practitioner: analyze data distributions before training and evaluate subgroup error rates.

Hour 1: Easy practice — bias autopsy

Scenario:

“A bank deploys an AI credit-approval model. It rejects loans for applicants from certain neighborhoods twice as often, despite identical income and credit scores.”

Tasks:

1. Identify the proxy variable causing geographic discrimination.
2. Explain why historical data caused this outcome.
3. Name two concrete harms suffered by applicants.
4. Suggest one immediate corrective measure.

Easy AI prompt:

“Analyze this loan approval case. Explain how postal code acted as a proxy variable for bias, and outline three technical steps to remediate geographic disparity.”

Hour 2: Legal regulations (GDPR & AI Act)

Theory: General Data Protection Regulation (GDPR)

- Right of access & erasure (Right to be forgotten).
- Article 22: Right not to be subject to automated decision-making producing significant legal effects without human intervention.
- Principle of purpose limitation and data minimization.

Theory: European Union AI Act (Risk-based approach)

- Unacceptable risk: prohibited (social scoring, subliminal manipulation, real-time biometric mass surveillance).
- High risk: strictly regulated (hiring tools, education grading, critical infrastructure, credit scoring).
- Limited/Minimal risk: transparency obligations (chatbots must self-identify as AI).

Legal obligations for high-risk AI:

- Rigorous data governance and quality management.
- Technical documentation and detailed logging (traceability).
- Transparency and provision of clear user instructions.
- Robust human oversight mechanisms built-in.
- High levels of cybersecurity, accuracy, and resilience.

Core principle:

Legal liability cannot be assigned to an AI algorithm; responsibility strictly rests on the deploying organization and provider.

Non-compliance risks: massive regulatory fines, operational bans, and reputational damage.

Hour 2: Easy practice — risk classification

Classify the following AI tools into EU AI Act risk tiers (Unacceptable, High, or Limited/Minimal):

1. AI used to screen job resumes for entry-level hiring.
2. AI government social scoring system rating citizen loyalty.
3. AI customer service chatbot answering product queries.
4. AI medical tool assisting doctors in tumor detection.
5. AI spam email filter.

Easy AI prompt:

“Explain why an AI resume-screening tool is classified as High Risk under the EU AI Act, and list three mandatory compliance requirements for its developers.”

Hour 3: Accountability & governance models

Theory: accountability definitions

- Accountability means having an identifiable human or entity answerable for the performance, ethics, and impacts of an AI system.
- Traceability: ability to inspect training data, model versions, and decision rationale.
- Explainability (XAI): providing understandable reasons why an AI reached a specific conclusion.

Accountability chain:

- Data providers: ensure data legality, consent, and cleanliness.
- AI developers: ensure fairness testing, security, and verification.
- Business managers: determine deployment boundaries and human oversight.
- End users/Operators: execute final human judgment.

Governance frameworks:

- Algorithmic impact assessments (AIA) before launch.
- Model documentation: model cards, datasheets for datasets.
- Incident reporting systems and clear redress procedures.

Human-in-the-loop vs. human-over-the-loop:

- In-the-loop: human actively approves every individual output.
- Over-the-loop: system operates autonomously while humans monitor dashboard alerts and retain override power.

Key takeaway: delegating a decision to AI does not delegate the legal or ethical liability.

Hour 3: Easy practice — incident role assignment

Scenario:

“An automated HR tool rejected a candidate due to a software hallucination in calculating years of experience. The candidate files an official complaint.”

Assign specific accountability tasks:

1. What must the AI engineer inspect?
2. What must the HR manager explain to the candidate?
3. What policy change should the leadership mandate?
4. How can human-in-the-loop prevent this in the future?

Easy AI prompt:

“Draft a standard operating procedure (SOP) ensuring human recruiters verify AI resume summaries before rejecting candidates.”

Hour 4: AI in public policy & societal governance

Theory: public sector AI considerations

- Public institutions deploy AI for welfare distribution, traffic control, tax fraud detection, and public healthcare allocation.
- Public sector AI requires higher standards of fairness, transparency, and public accountability than commercial apps.
- Digital divide: AI services must not disenfranchise citizens without digital literacy or high-speed internet.

Risks in governance:

- Surveillance creep and erosion of civil liberties.
- Opacity: citizens unable to challenge "black-box" government decisions (e.g., Dutch SyRI welfare fraud scandal).
- Reinforcing socioeconomic inequalities through biased predictive policing.

Public policy principles for AI:

- Open algorithms & public algorithmic registries.
- Democratic participation: public consultation before deploying civic AI.
- Universal accessibility and non-discrimination.
- Independent regulatory oversight and external audits.

Case example (Dutch Childcare Benefits Scandal):

Algorithms flagged dual-nationality parents for fraud audits based on discriminatory proxies, leading to severe injustice and government resignation.

Hour 4: Easy practice — policy audit critique

Scenario:

“A city council proposes using AI cameras to automatically fine citizens for littering, with fines sent directly without human police verification.”

Tasks:

1. Identify two civil rights or fairness concerns.
2. Identify one major technical failure mode (false positive).
3. Propose two policy safeguards to protect citizens.
4. Write an appeal process requirement.

Expected points:

- Concerns: mass biometric surveillance, lack of due process.
- Failure mode: mistaking dropping an object by accident for intentional littering.
- Safeguards: require human officer review before issuing fines; ban facial recognition for minor infractions.
- Appeal process: free, fast online and in-person review mechanism.

Easy AI prompt:

“Critique this automated civic fine proposal from a public policy perspective. Identify three failure modes and outline a fair citizen appeal mechanism.”

Hour 5: Data privacy, security & confidentiality

Theory: data privacy risks with AI

• Personally Identifiable Information (PII): names, NIDs, emails, phone numbers, health records, biometric signatures.

• Corporate confidentiality: proprietary code, financial forecasts, trade secrets, confidential client briefs.

• **Data leakage mechanisms:**

- Ingestion leakage: public LLMs using input prompts for future model training.

- Memorization leakage: generative models reciting sensitive training data verbatim.

- Prompt injection & jailbreaks: bypassing safety guards to reveal backend system data.

Security fundamentals:

• Zero Data Retention (ZDR) enterprise APIs vs. public free AI chat tools.

• Differential privacy and data masking techniques.

• Secure environment sandboxing (preventing model data exfiltration).

Safety rule for employees:

Never paste unredacted PII, passwords, internal financial tables, or customer records into public AI consumer tools (ChatGPT free tier, Copilot personal, etc.).

Hour 5: Easy practice — PII sanitization lab

Raw text snippet:

“Client John Doe (NID: 1984-562910, phone: +8801711000000) deposited 500,000 BDT into Account #99281. He requested a portfolio review.”

Tasks:

1. Identify all 5 sensitive PII elements.
2. Rewrite the snippet into a fully anonymized version suitable for AI prompting.
3. Formulate the safe prompt for market research.

Easy AI prompt:

“Act as a financial analyst. Based on a client with a Tier-3 deposit seeking moderate growth, provide three asset-allocation strategies. (No real customer data provided).”

Reflection: Why does anonymization preserve AI utility while eliminating privacy risk?

Hour 6: Data protection best practices & wrap-up

Theory: organizational best practices

- Data minimization: send only the necessary summarized context to the AI.
- Synthetic data generation: test AI workflows using artificial dummy data.
- Clear internal AI usage policies: approved tool lists, classification levels.
- Continuous training: educating employees on social engineering and data leakage.

6-Point AI safety checklist before hitting send:

1. Is all personal/customer PII stripped or masked?
2. Is company confidential information protected?
3. Is the AI tool enterprise-approved with privacy protections?
4. Did I verify all factual claims and calculations?
5. Are the criteria fair and free of biased proxies?
6. Does a human maintain accountability for the final output?

Day 19 summary & key takeaways:

- Bias in AI stems from human history, skewed data, and poor metric selection.
- GDPR and EU AI Act establish legal boundaries and mandate human oversight for high-risk applications.
- Accountability always stays with human practitioners and organizations.
- Privacy and security require proactive anonymization and data hygiene.

Exit ticket (5 minutes):

1. Name one high-risk AI application under EU law.
2. List two types of PII you must never feed into public AI.
3. State one concrete habit you will follow to ensure responsible AI use.